

Financial NLP and Sentiment Analysis

Lecture 09 — Treating Language as Financial Data

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June 20, 2026

Outline

- 1 Text as Financial Data
- 2 Sentiment Analysis Methods
- 3 Applications by Text Source
- 4 From Sentiment to Tradeable Signal
- 5 Practical Pipeline

Why Treat Language as Data?

THE BIGGER PICTURE

A single trading day now generates hundreds of thousands of news articles, social posts, filings, and analyst reports. Extracting the signal that moves prices — in real time, repeatably — is a defining problem of quantitative finance this decade.

Text is a data source with its own **measurement challenges**, exactly as:

- price data has bid–ask bounce,
- volume data has reporting lags,
- **text** has selection, duplication, and timestamp pitfalls.

This lecture

Where financial text comes from → how to score its sentiment → what each source predicts → how to turn scores into a tradeable signal → the production pipeline.

Five Sources of Financial Text

Sources differ on four axes: **frequency**, **curation**, **audience**, **regulatory oversight**.

Source	Curation / latency	Canonical finding
Social media	none / real-time	Bollen 2011: Twitter mood → DJIA
News & wires	high / ms-hours	Tetlock 2008: neg. words → returns
Earnings calls	scripted + Q&A	Larcker 2012: tone → restatements
SEC filings	legal / quarterly	Loughran-McDonald 2011
Central banks	extreme / scheduled	Schmeling 2019: tone → equities

Source shapes method

The structural properties of each source determine both the information it carries *and* the right preprocessing / model. There is no one-size-fits-all pipeline.

Source Properties That Matter

Social media (X, Reddit/WSB, StockTwits)

- Least curated; high noise, sarcasm, bots
- Captures *retail attention* orthogonal to institutions
- Aggregates inform even when posts don't

News & wires

- Reuters/Bloomberg: ms latency, structured event
- Key flaw: **selection** — only newsworthy firms

Earnings calls

- Prepared remarks = soft-info framing
- Q&A reveals analyst concerns + candour

SEC filings (EDGAR): 10-K (Item 1A, MD&A), 10-Q, 8-K, DEF 14A

- Disclosure law $\Rightarrow$ truthful but vague

Central banks: FOMC, ECB

- Meaning in calibrated *deviations* from boilerplate

Preprocessing: From Raw Text to Tokens

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Every pipeline normalises, deduplicates, and time-aligns before modelling. Skipping any of these biases the downstream sentiment measure.

Tokenization & normalization

- Subword tokenizers: BPE (GPT) / WordPiece (BERT).
- Case folding for lexicons; *preserve* case for cased transformers.
- Number handling ([NUM]) — but EPS \$2.40 vs \$0.24 carries signal.
- Cashtag/ticker harmonisation (\$AAPL); boilerplate removal (XBRL, exhibits).

Deduplication — most important, most neglected

- Wires reprint verbatim; retweets propagate identical text.
- Duplicates bias sentiment *and* inflate sample size → spurious narrow CIs.
- MinHash / LSH; mark dupes when trigram Jaccard > 0.6–0.8.

Temporal alignment — avoid **look-ahead bias** (post-close news → same-day return) and **stale timestamps**; store publication *and* retrieval times separately

Three Coexisting Generations

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No single method dominates. Lexicons are fast and interpretable; fine-tuned classifiers are strong in-domain; LLMs are flexible and need no labels. A competent practitioner uses all three and knows when each fits.

Method	Labels?	Strength	Weakness
Lexicon (LM)	none	transparent, cheap	no negation/context
Fine-tuned (FinBERT)	yes	strong in-domain	needs labelled data
LLM zero/few-shot	none	flexible, nuanced	cost & latency

Lexicon Method: Loughran–McDonald

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Net sentiment ratio for document d :

$$S_{\text{LM}}(d) = \frac{N_{\text{pos}}(d) - N_{\text{neg}}(d)}{N_{\text{words}}(d)}$$

Why a finance dictionary? Harvard General Inquirer mislabels *tax*, *liability*, *cost*, *foreign*, *crude*, *cancer* as negative — neutral in 10-Ks. On 10-Ks, GI negativity tracks legal boilerplate, not performance.

LM word lists (built from the 10-K vocabulary):

Negative	Litigious	Positive	Uncertainty	Weak modal	Strong modal
2,355	903	354	297	27	19

The **negative** list dominates — the asymmetry of disclosure language.

LM Worked Example & Limits

MD&A excerpt

“Revenue declined 12%, reflecting challenging conditions and competitive pressure. The Company remains cautious but is confident in its long-term position.”

- Negative hits: *declined, challenging, pressure, cautious* = 4
- Positive hits: *confident* = 1; total non-trivial words ≈ 35
- $S_{LM} = (1 - 4)/35 \approx -0.086$ (bottom-quartile MD&A)
- A GI classifier also flags *competitive* $\rightarrow \approx -0.17$: a distortion.

Known limits

No negation (“not confident” scores positive); context-blind; domain shift (misses “headwinds”, “softness”). Still widely used: transparent, cheap, auditable.

Fine-Tuned Classifiers: FinBERT & SEC-BERT

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Take a pre-trained encoder, attach a head, fine-tune. For the [CLS] embedding $\mathbf{h} \in \mathbb{R}^{d_h}$:

$$P(y = c \mid d) = \text{softmax}(\mathbf{W}_c \mathbf{h} + \mathbf{b}_c), \quad c \in \{\text{pos, neu, neg}\}$$

minimising cross-entropy; predict $\hat{y} = \arg \max_c P(y = c \mid d)$.

FinBERT (Araci 2019; Huang 2023) SEC-BERT (Loukas 2022)

- Further-pretrain BERT on financial news + filings
 - Fine-tune on PhraseBank + analyst + call data
 - Beats GI, LM, and generic BERT
- Pre-trained *only* on EDGAR (10-K/Q/8-K)
 - Vocabulary tuned to legal-financial text
 - Wins in-domain on risk-factor / event tasks

Edge over lexicons: $\mathbf{h}$ sees the whole sequence $\Rightarrow$ handles negation,

The Financial PhraseBank

The standard labelled benchmark (Malo et al. 2014):

- **4,840** news sentences, labelled pos / neu / neg.
- **16** finance-trained annotators; majority label used.
- At 75% agreement: $\approx 3,400$ high-confidence sentences, inter-annotator agreement $\approx 75\text{--}80\%$.
- **Neutral $\approx$ half the corpus** — motivates macro-F1, not accuracy.

Caveat

News-sentence style $\neq$ filings / call language. Models tuned on PhraseBank may transfer poorly to long, syntactically complex documents — supplement with domain-matched labels.

LLM Zero-Shot & Few-Shot Classification

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GPT-4 / Claude / LLaMA-3 classify sentiment from a task description alone — *no fine-tuning, no labelled set*. Rapid prototyping on new domains.

Zero-shot prompt sketch: “*You are a financial analyst. Classify the sentiment . . . from the perspective of a stock investor.*” → “Neutral” with a rationale (revenue growth offset by margin compression).

Benchmarks (real findings):

- Ko & Lee 2024: GPT-4 zero-shot *matches or exceeds* FinBERT on PhraseBank, FiQA, and a proprietary call dataset — with no labels.
- López-Lira & Tang 2023: GPT headline scores positively predict next-day returns, beating lexicons.
- Kirtac & Germano 2024: OPT (GPT-3 family) yields a markedly higher long–short Sharpe than the LM dictionary.
- Fatemi & Hu 2023: fine-tuned *small* LLMs hit SOTA; extra few-shot examples beyond a small set add no accuracy.

One Source, One Research Question

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Each source has a natural question: does retail mood drive short-run prices? does news carry information beyond prices? does management tone reveal private information? does disclosure language predict performance? do policy words move asset prices?

Source	Question
Social media	Retail attention → price pressure?
News	Incremental information beyond prices?
Earnings calls	Tone → private information?
SEC filings	Disclosure language → future risk/return?
Central banks	Linguistic cues → asset prices?

Social Media: Attention, Disagreement, Bank Runs

Evidence:

- Bollen 2011: “calm” Twitter mood Granger-causes DJIA (10-week sample; mixed replications).
- Cookson & Niessner 2020: StockTwits *disagreement* predicts volume and spreads.
- Gil-Bazo et al. 2025: 1.6M fund-family posts — tone *and* volume predict fund inflows.
- Cookson et al. 2026: SVB collapse — high Twitter-exposure banks lost more value; social media can *catalyse* bank runs (streaming use case).

WSB / GameStop (Jan 2021): coordinated WSB sentiment led GME/AMC by 1–2 days; mechanism via call demand → gamma-hedging amplification.

Methodological hazards

Bots & coordinated inauthentic behaviour; severe selection; reverse causality (returns → posting).

News (Tetlock 2007/08)

- WSJ pessimism → price pressure + reversion
- Firm-level neg. language → earnings
- Stale negativity adds no forecast power

Earnings calls

- Larcker 2012: deception cues → restatements
- Price 2012: tone → post-call abnormal returns
- Cook & Kazinnik 2023: local LLMs ≈ closed-source, lower privacy risk

SEC filings

- LM negativity ↓ filing-day return, ↑ volume
- Cohen 2020 “lazy prices”: unchanged 10-Ks underperform
- Lehner 2024: LLMs can rewrite filings to shift measured tone

Central banks

- Schmeling 2019: positive tone → equities ↑
- Hansen 2018: deliberation quality → inflation outcomes
- Nakamura 2018: **information vs. policy** effect confound

MD&A Readability: The Fog Index

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$$\text{Fog} = 0.4 \left(\frac{N_{\text{words}}}{N_{\text{sentences}}} + 100 \cdot \frac{N_{\text{complex}}}{N_{\text{words}}} \right)$$

N_{complex} = words with ≥ 3 syllables (excl. proper nouns, compound modifiers, simple verb inflections).

- Fog = 12 $\Leftrightarrow$ US high-school level.
- 10-K MD&A typically **18–22** — much harder than a newspaper.
- Li 2008: higher Fog $\rightarrow$ lower current earnings, higher earnings volatility (managers obscure when performance is poor).
- Lehavy 2011: less readable 10-Ks $\rightarrow$ fewer, less accurate analysts.

Raw Scores Are Not Yet a Signal

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A score per document is not tradeable. It needs aggregation, denoising, and statistical transformation. Two empirical frameworks test economic content: event studies and predictive regressions.

Document → **entity aggregation** (all docs on firm i , day t):

- Simple mean: $\bar{S}_t = \frac{1}{n_t} \sum_i S_i$
- Volume-weighted (reach / circulation) · Recency-weighted (exp. decay)

Noisy sources (social) need weighting; curated sources (filings) do not.

Standardization & Stationarity

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Cross-sectional z-score within industry-year removes sector language fixed effects:

$$\tilde{S}_{i,t} = \frac{S_{i,t} - \mu_{j(i),t}}{\sigma_{j(i),t}}$$

($j(i)$ = industry of firm i .) A defence firm and a software firm become comparable.

Stationarity

- Long sentiment series drift (style change, universe change, regime shift).
- Apply ADF / KPSS; first-difference or detrend non-stationary series.

Sentiment inflation

Re-scoring old text with a newer model changes measured sentiment. Use one *fixed* model vintage across all periods — or document the vintages.

UNDER THE HOOD

Abnormal return $AR_{i,\tau} = R_{i,\tau} - \hat{R}_{i,\tau}$ (benchmark, e.g. CAPM). Cumulative over $[a, b]$:

$$CAR_i[a, b] = \sum_{\tau=a}^b AR_{i,\tau}, \quad \overline{CAR}[a, b] = \frac{1}{N} \sum_i CAR_i[a, b].$$

Does text beat the EPS number? Condition on the earnings surprise:

$$CAR_i[1, 60] = \alpha + \beta_1 SUE_{i,0} + \beta_2 \tilde{S}_{i,0} + \varepsilon_{i,0}$$

with $SUE = (EPS - \widehat{EPS})/\hat{\sigma}$. Incremental text content $\Rightarrow \hat{\beta}_2 > 0$.

High-frequency FOMC: 30-min window around announcements (Nakamura 2018) demands sentence-level scoring *inside* the window $\rightarrow$ low-latency inference.

Two Horizons, Two Mechanisms

	Short horizon	Long horizon
Channel	attention / price pressure	fundamental information
Evidence	Da 2015: attention → reversal	Tetlock 2008; Cohen 2020
Data needs	real-time, low-latency	daily/weekly, heavier models
Audience	HFT-adjacent industry	most academic research
Alpha	larger (pre-cost)	smaller, more accessible

Practical implication

Short-horizon strategies need real-time feeds and execution within minutes; long-horizon strategies tolerate batch processing and richer models. Industry chases the short horizon (more alpha); academia studies the long horizon (lower data/execution cost).

From “What” to “How”

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Earlier sections said what to extract and why. The pipeline is the engineering: collect text at scale, run inference in production, evaluate classifier quality with metrics that don't mislead.

Three production concerns:

- ➊ **Data collection** — EDGAR, social APIs, news vendors.
- ➋ **Inference at scale** — batch (throughput) vs. streaming (latency).
- ➌ **Evaluation** — label quality *and* classifier quality *and* economic validity.

(Collection & scale engineering detailed in the appendix.)

Evaluation: Annotator Agreement

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Krippendorff's alpha measures agreement beyond chance:

$$\alpha = 1 - \frac{D_o}{D_e}$$

D_o = observed, D_e = expected disagreement. Nominal k -class: $D_e = 1 - \sum_c (n_c/n)^2$. $\alpha = 0$: chance; $\alpha = 1$: perfect.

Thresholds: $\alpha \geq 0.667$ reliable; ≥ 0.800 high confidence. Financial benchmarks: **0.65–0.78** — genuine ambiguity of finance sentences.

Dataset	Classes	Alpha
Financial PhraseBank (100% agr.)	3	1.00
Financial PhraseBank (66% agr.)	3	≈ 0.72
Earnings call sentences (practitioner)	3	0.68–0.75
FOMC statement sentences	3	0.70–0.78

Evaluation: Classifier Metrics & Economic Validity

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Per class c :

$$P_c = \frac{TP_c}{TP_c + FP_c}, R_c = \frac{TP_c}{TP_c + FN_c}, F1_c = \frac{2P_cR_c}{P_c + R_c}.$$

Report **macro-F1** (unweighted mean over classes).

Why macro, not accuracy

Neutral $\approx$ half of financial sentences. An accuracy-maximiser that over-predicts neutral scores high while failing on the economically important positive / negative classes.

Economic validity $\neq$ statistical metrics. Macro-F1 = 0.75 can still have zero alpha if the test set is unrepresentative or labels are misaligned. Best practice: regress aggregate sentiment on forward returns / earnings

Lecture 09 Summary

- **Sources:** social → filings, spanning curation and oversight; structure dictates preprocessing and model.
- **Methods:** lexicon (LM, the filings standard), fine-tuned (FinBERT/SEC-BERT), LLM zero/few-shot — LLMs match or beat fine-tuned without labels, at higher cost.
- **Applications:** attention (social), information (news), soft disclosure (calls), risk (filings), policy signal (central banks).
- **Signal:** aggregate → z-score within industry-year → test stationarity; event study + predictive regression assess economic content.
- **Pipeline:** batch + streaming inference; evaluate with Krippendorff's α , macro-F1, *and* a held-out economic test.

Next

Practical 09 — LM scoring, FinBERT vs. lexicon, and a filing-date event study on real EDGAR + price data.

APPENDIX

Data Collection, Inference at Scale, and Extensions

Extra / optional material — beyond the core lecture.

EDGAR full-text search (`efits.sec.gov`)

- 1 Query form types (10-K, 8-K) and date range.
- 2 Download index pages → document links.
- 3 Parse primary HTML/SGML (BeautifulSoup, `sec-edgar-downloader`).
- 4 Extract MD&A / Risk Factors by header matching.
- 5 Store Parquet, indexed by CIK / period / filing date.

Twitter/X Academic API (paid since 2022): filters `has:cashtags`, `lang:en`, `verified-author`. Alternatives: StockTwits API, Reddit Pushshift (restricted), commercial sentiment feeds.

News vendors: Bloomberg, Dow Jones DNA, Refinitiv, RavenPack (machine-readable event records); GDELT (free, noisier, 15-min updates) for research.

Batch (throughput-bound):

- FinBERT (110M), 512 tokens, single A100: hundreds of docs/sec; multi-million archive in hours.
- Optimisations: dynamic padding/batching; FP16/BF16; distillation (110M → 33M, < 5% accuracy loss); data/CPU parallelism.

Streaming (latency-bound):

- Target < 100 ms for FOMC; near-real-time for live call transcription.
- Kafka/Kinesis → ONNX inference workers → downstream aggregation.
- `onnxruntime` + quantised weights: < 100 ms/sentence on modern CPUs.

Appendix: Ordinal Krippendorff's Alpha

For ordered labels (pos > neu > neg) the nominal coincidence metric is replaced by a weighted disagreement:

$$d^2(c, c') = \left(\sum_{g=c}^{c'} n_g \right)^2,$$

penalising disagreements by the number of scale steps between labels (n_g = frequency of category g). Ordinal α is more conservative than nominal α for sentiment corpora — a stricter lower bound on agreement.

News-Implied Volatility (Manela & Moreira 2017): project WSJ front-page text onto realized VIX variation; the measure extends to **1890**, capturing disaster-risk eras (Depression, WWII, 2008–09) far beyond VIX's 1990 start.

News in factor models (Ke et al. 2019): supervised extraction of price-relevant news features adds predictive content, esp. at short horizons; intraday alpha captured by HFT, longer-horizon signals lower-Sharpe but accessible.

Climate / ESG (Engle et al. 2020): a climate-news index builds a hedge portfolio; high climate-news exposure → lower returns on high-news days. Generalises to pandemic / geopolitical / supply-chain risk indices.

Beyond the context window:

- Chiu & Hung 2024: finance-specific LLaMA-2 + LLM summarisation of long 10-Ks → higher buy-and-hold abnormal returns than FinBERT.
- Siano 2025: LLM word-context embeddings of earnings press releases explain far more return variation than dictionaries; raise R^2 of financial-statement surprises.
- Xu & Babaian 2025: FinBERT + GPT-4o *ensemble* on FOMC press conferences — cross-family combination is more stable than any single model.